A First Course in Abstract Algebra

Section 21: The Field of Quotients of an Integral Domain

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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21 The Field of Quotients of an Integral Domain

Exercise 21.1: Quotients of Gaussian integers

The field is $\boxed{\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\}}$. To see that every quotient lies here, rationalize its denominator:

$$\frac{m + ni}{r + si} = \frac{(mr + ns) + (nr - ms)i}{r^2 + s^2}, \qquad r + si \neq 0.$$

Both resulting coefficients are rational. Conversely, if $a, b \in \mathbb{Q}$, choose a common nonzero integer denominator d with $da, db \in \mathbb{Z}$. Then $a + bi = ((da) + (db)i)/d$ is a quotient of Gaussian integers. These two inclusions establish the description.

Exercise 21.2: Quotients with an irrational square root

The answer is $\boxed{\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}}$. A quotient rationalizes as

$$\frac{m + n\sqrt{2}}{r + s\sqrt{2}} = \frac{(mr - 2ns) + (nr - ms)\sqrt{2}}{r^2 - 2s^2}.$$

The denominator is nonzero whenever $r + s\sqrt{2} \neq 0$: otherwise irrationality of $\sqrt{2}$ forces $r = s = 0$. Thus the quotient has rational coefficients. Conversely, clearing the two rational denominators expresses any $a + b\sqrt{2}$ as a quotient of two members of $\mathbb{Z}[\sqrt{2}]$.

Exercise 21.3: Correcting the field-of-quotients definition

The missing condition is that *every element of* F be a quotient of elements of the embedded copy of D. Precisely, a field of quotients of D is a field F with an injective ring map $i : D \to F$ such that every $x \in F$ is $i(a)i(b)^{-1}$ for some $a, b \in D$, $b \neq 0$. Merely making all nonzero elements of D invertible allows larger fields such as $\mathbb{R}$ for $D = \mathbb{Z}$, and does not characterize the field of quotients.

Exercise 21.4: True or false: fraction fields

(a) True: every rational number is a quotient of integers.

(b) False: an irrational real number cannot be a quotient of integers.

(c) True: every real r is the quotient $r/1$ of real numbers.

(d) False: a quotient of real numbers is real, so it cannot equal i.

(e) True: if D is already a field, all its fractions $a/b = ab^{-1}$ remain in D.

(f) True: nonzero denominators must have nonzero products, and cancellation is used to establish transitivity of the equivalence relation.

(g) False: zero is an element of D and has no multiplicative inverse in a field.

(h) True: the image of a nonzero b has inverse represented by the fraction $1/b$.

(i) True: in a fraction field of D, the fractions a/b with $a, b \in D'$ form a subfield isomorphic to the fraction field of D'. Equality is the same cross-product equality in both domains.

(j) True: uniqueness of the fraction field up to isomorphism applies to $\mathbb{Z}$, and $\mathbb{Q}$ is one such field.

Exercise 21.5: Proper subdomains with the same fraction field

Take $D' = \mathbb{Z}$ and $D = \mathbb{Z}[1/2] = \{m/2^k : m \in \mathbb{Z},\ k \geq 0\}$. These are integral domains and D' is proper because $1/2 \in D \setminus D'$. The fraction field of either is $\mathbb{Q}$: fractions of elements of D are rational, and every rational is already a quotient of integers. Thus $F' = \mathbb{Q}$ works for both.

Exercise 21.6: Associativity of fraction addition

Write $[a, b]$ for the class $[(a, b)]$, and let $b, d, f \neq 0$. The addition formula gives

$$([a, b] + [c, d]) + [e, f] = [(ad + bc)f + (bd)e, bdf],$$

whereas

$$[a, b] + ([c, d] + [e, f]) = [a(df) + b(cf + de), bdf].$$

The numerators are both $adf + bcf + bde$ by the distributive and associative laws in D. The resulting ordered pairs are identical, hence so are their classes. This proves Part 2 of Step 3.

Exercise 21.7: The additive identity in the fraction construction

Using the prescribed operations,

$$[a, b] + [0, 1] = [a \cdot 1 + b \cdot 0, b \cdot 1] = [a, b].$$

Commutativity of addition, established in Part 1, also gives $[0, 1] + [a, b] = [a, b]$. Thus $[0, 1]$ is an additive identity, proving Part 3.

Exercise 21.8: The additive inverse of a fraction

Since $b \neq 0$, its square is nonzero, and $[-a, b]$ is an admissible fraction. Then

$$[a, b] + [-a, b] = [ab + b(-a), b^2] = [0, b^2] = [0, 1].$$

The last equality uses the defining equivalence: $0 \cdot 1 = b^2 \cdot 0$. Commutative addition supplies the equation in the other order. Hence $[-a, b]$ is the additive inverse, proving Part 4.

Exercise 21.9: Associativity of fraction multiplication

For three fractions, the multiplication rule gives

$$([a, b][c, d])[e, f] = [(ac)e, (bd)f], \qquad [a, b]([c, d][e, f]) = [a(ce), b(df)].$$

Associativity in D makes both numerators and both denominators identical. Nonzero denominator products follow from the integral-domain hypothesis. Therefore the classes are equal, proving Part 5.

Exercise 21.10: Commutativity of fraction multiplication

The products are $[a,b][c,d] = [ac, bd]$ and $[c,d][a,b] = [ca, db]$. Both coordinates agree because D is commutative. Thus the products are equal, which proves Part 6.

Exercise 21.11: Distributivity in the fraction construction

Let $x = [a,b]$, $y = [c,d]$, $z = [e,f]$. The prescribed formulas give

$$x(y+z) = [a(cf+de), bdf], \qquad xy + xz = [acbf + abde, b^2df].$$

The latter numerator is $b\,a(cf+de)$, so the two fractions are equivalent: their cross products are both $a(cf+de)b^2df$. Thus $x(y+z) = xy + xz$. Multiplication has already been proved commutative in Part 6, so

$$(y+z)x = x(y+z) = xy + xz = yx + zx.$$

This proves both distributive laws, Part 7.

Exercise 21.12: Partial quotients and a unity

Use classes $[r,t]$ of pairs in $R \times T$, with $[r,t] = [s,u]$ when $ru = st$. Transitivity follows by canceling a denominator from T; cancellation is valid because its elements are nonzero and not zero divisors. Products of denominators remain in T. The ordinary fraction addition and multiplication formulas are well defined by cross multiplication, just as in the text. They give a commutative ring by the same coordinate expansions as Exercises 21.6–21.11.

To view R inside it, fix any $t \in T$ and set $i(r) = [rt, t]$. The class is independent of t, because $(rt)u = (ru)t$. The formulas show that i preserves sums and products. If $i(r) = i(s)$, then $(r-s)t^2 = 0$; two cancellations of the regular element t give $r = s$. Thus i is an embedding.

(a) The element $e = [t,t]$ is independent of $t \in T$. It is a unity, since $[t,t][r,u] = [tr, tu] = [r,u]$ by cross multiplication. No unity in R was needed.

(b) For $u \in T$ and $t \in T$, the element $i(u) = [ut, t]$ has inverse $[t, ut]$, whose denominator lies in T. Their product is $[ut^2, ut^2] = e$. Hence every embedded element of T is a unit.

Exercise 21.13: Adjoining unity using regular powers

For a *nonzero* element a that is not a zero divisor, let $T = \{a^n : n \geq 1\}$. It is nonempty and multiplicatively closed. None of its elements is zero or a zero divisor: an equation $a^n x = 0$ permits n successive cancellations of a and forces $x = 0$. Exercise 21.12 therefore embeds R in the commutative ring $Q(R,T)$ with unity.

The qualification $a \neq 0$ is necessary for this particular argument: under the book's convention zero itself is not called a zero divisor, but its powers cannot serve as denominators. If the printed hypothesis is read as allowing only $a = 0$, use the general unity-adjoining construction of Exercise 19.30 instead. That construction applies to every ring, whereas localization additionally makes the chosen regular denominators invertible.

Exercise 21.14: Partial quotients of the ring modulo four

The allowed denominators $1, 3$ are already units in $\mathbb{Z}_4$, with $3^{-1} = 3$. Send $[r, t]$ to $rt^{-1} \in \mathbb{Z}_4$. Cross multiplication shows this is well defined and injective, and every residue occurs as $[r, 1]$. It is a ring isomorphism. Thus the partial-quotient ring has $\boxed{4}$ elements.

Exercise 21.15: Inverting powers of two

Send a formal fraction $[m, 2^n]$ to the real rational $m/2^n$. Cross multiplication makes this a well-defined injective ring map. Its image is

$$\boxed{\mathbb{Z}[1/2] = \{m/2^n : m \in \mathbb{Z},\ n \geq 0\}}.$$

Allowing $n = 0$ does not enlarge the image: $m = (2m)/2$. These are the dyadic rationals; denominators may contain only the prime 2 after reduction.

Exercise 21.16: Quotients of multiples of three

The map $[3m, 6^n] \mapsto 3m/6^n$ identifies the required ring with a subring of $\mathbb{Q}$. Every such fraction has denominator involving only 2 and 3, so its image is contained in $\mathbb{Z}[1/6]$. Conversely any $k/6^j$, $j \geq 0$, equals $6k/6^{j+1}$; the numerator $6k$ lies in $3\mathbb{Z}$ and the denominator is an allowed positive power. Therefore the image is exactly

$$\boxed{\mathbb{Z}[1/6] = \{k/6^j : k \in \mathbb{Z},\ j \geq 0\}}.$$

Exercise 21.17: The failure when denominators are zero divisors

The cross-product relation need not be transitive. In $R = \mathbb{Z}_6$, with $T = \{1, 2, 4\}$, consider

$$(0, 1), \qquad (0, 2), \qquad (3, 1).$$

The first two are related because $0 \cdot 2 = 1 \cdot 0$. The last two are related because $0 \cdot 1 = 2 \cdot 3 = 0$ modulo 6. But the first and third are not related, because $0 \cdot 1 = 0 \neq 3 = 1 \cdot 3$. Thus Step 1 already fails to define an equivalence relation. The transitivity proof would have canceled 2, which is a zero divisor. More general localization can be defined by a different relation, but then elements annihilated by denominators collapse to zero and the original ring need not embed.